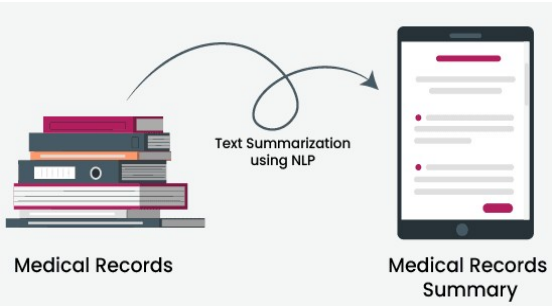
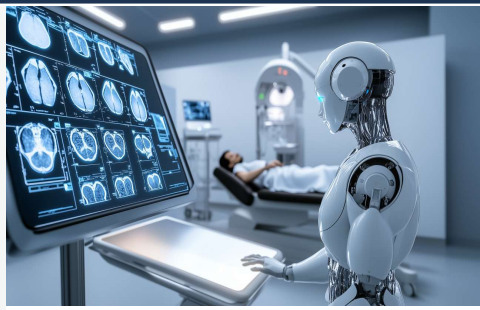
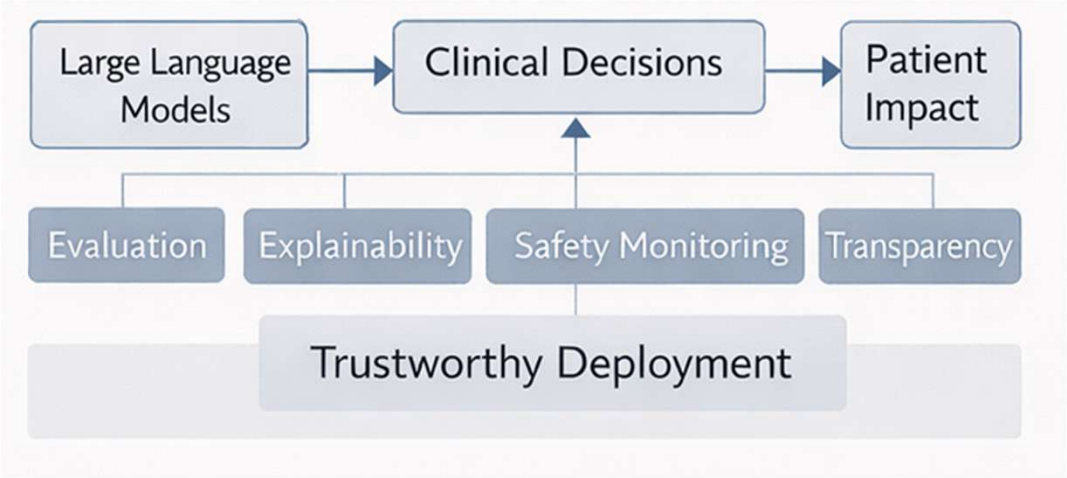
Trustworthy And Safe AI For Clinical Systems

Motivation

- Large language models increasingly support clinical summarization, diagnosis, and decisions.
- Current systems remains opaque, prone to hallucinated reasoning.
- Safety risks arise from unreliable outputs, inconsistent reasoning, and unverifiable decisions.
- Clinical settings require transparency, traceability, and explicit failure analysis.







Goal

Build reliable, explainable, and traceable AI systems that are safe enough to deploy.

Trustworthy And Safe AI For Clinical Systems

Why Trustworthiness matters

- Clinical AI operates in high-risk environments with direct patient impact.
- Model outputs must be fair, reliable, interpretable, and verifiable.
- Incorrect reasoning with correct predictions remains unsafe.
- Deployment requires safeguards against hallucinations and failures.

Trustworthy AI: Focus Areas

- Reliability: Clinical AI must operate on accurate and complete clinical facts.
- Safety: Prevention of harmful or unsafe recommendations.
- Explainability: Interpretable reasoning and decision traceability.
- Fairness: Mitigate Bias and improve Diagnostic equity

Deployable Clinical AI

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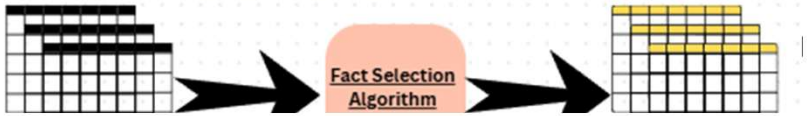
Key challenges:

Reliable Clinical Information:

- Clinical AI must operate on accurate and complete clinical facts.

Consistent Clinical Understanding:

- System ... ages and contexts.



Safe Clinic ...

- AI mus ... nent recommendations.

Language	Example Q–A Pair
English	Q: What is the diagnosis? A: Throat infection with mild laryngitis.
Hindi	Q: Rogi ki mukhya shikayat kya hai? A: Pichhle do mahine se gale mein kharash aur jalan.
Tamil	Q: Noyaliyin parisothanai mudivugal enna? A: CT scan kural kuruthil veekkam kaattugi-rathu.

Measurab ...

- Trust ir ... , and auditable.

Slide 6

AS1 need to add animations in slide 6 with examples
Amit Sangroya, 2026-03-16T08:49:32.523

Agenda

- Motivation
- Reliable CSR Summarization
- Safe and Traceable Medical Diagnosis
  - Med-Guard & Trust-X
- Equitable Dermatology
  - Fairness, Bias
- Explainable Diagnosis
  - Derm-Clu
- Summary

Structured Fact Extraction for Reliable Clinical Table Summarization

Problem

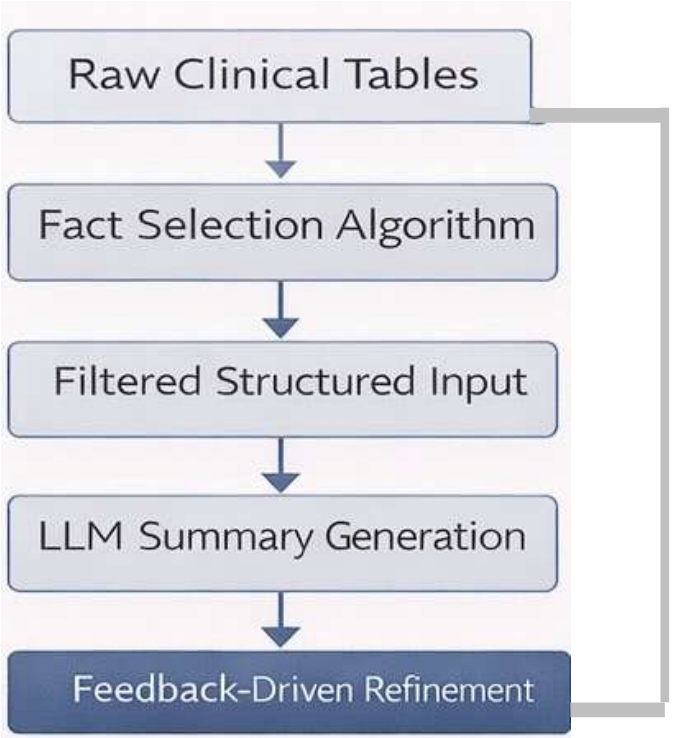
- Clinical study reports contain large structured tables with dense statistical and medical information.
- Direct prompting of LLMs produces incomplete or inconsistent summaries due to context length limits.
- Models frequently omit critical facts or generate imprecise clinical statements.

Key Idea

- Decompose table-to-text generation into fact selection and controlled generation.
- Extract clinically relevant information prior to summary generation.
- Reduce input complexity to improve factual consistency and precision.

Contribution

- End-to-end pipeline integrating fact selection and feedback loop
- Improves factual completeness and reduces hallucination content.
- Enables more reliable and clinically usable summaries.



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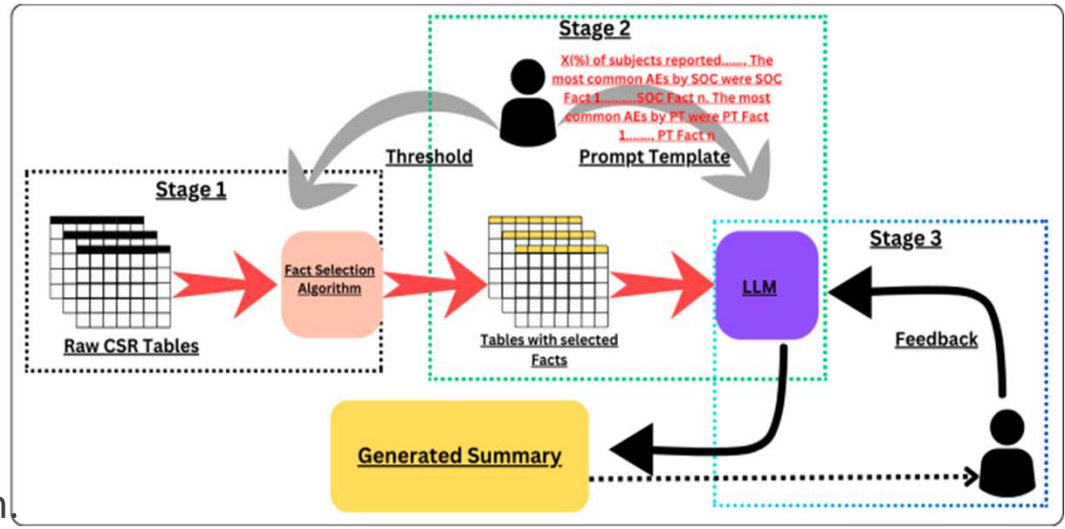
Method: CSR Table Summarization Pipeline

Task

- Generate summaries from large clinical trial tables.
- Preserve and extract all important clinical facts.
- Avoid hallucinations.
- Integrate a feedback loop, allowing users to refine the generated summaries.



Pipeline Steps

- **Table Linearization:** Convert structured tables into sequence format for LLM input.
- **Fact Selection:** Extract important facts using algorithm.
- **Input Reduction:** Remove irrelevant or low-frequency data.
- **Summary Gen.:** LLM generates structured clinical summary.
- **Feedback Loop:** Users refine output for accuracy and clarity.

Results: CSR Table Summarization Pipeline

Evaluation

- Used Claim Recall (completeness) and Claim Precision (conciseness).
- Used GPT-4o as LLM as a judge.
- Human evaluation using internal medical writer.
- Checked rule compliance, accuracy, conciseness, fluency.

Quantitative Results

- **GPT-4o-mini:**
  - Claim Recall: 0.67
  - Claim Precision: 0.47
- **Nous-Hermes-2-Mixtral (**with fact selection algorithm**):**
  - Claim Recall: 0.72
  - Claim Precision: 0.44

Table 4: Comparison of Claim Recall and Precision Across Different Models and Approaches

Model	Claim Recall	Claim Precision
Nous-Hermes-2-Mixtral-8x7B DPO (with fact selection algorithm) **Our Approach**	**0.72**	**0.44**
GPT 4o-mini (0-shot)	0.58	0.38
GPT 4o-mini (1-shot)	0.67	0.47
DeepSeek (0-shot)	0.5	0.36
DeepSeek (1-shot)	0.55	0.44
Llama-3.1-70B-Instruct (0-shot)	0.18	0.15
Llama-3.1-70B-Instruct (1-shot)	0.22	0.18
Nous-Hermes-2-Mixtral-8x7B DPO (0-shot)	0.27	0.22
Nous-Hermes-2-Mixtral-8x7B DPO (1-shot)	0.23	0.29

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Med-Guard: Safety First Multi-Agent Clinical Decision System

Problem

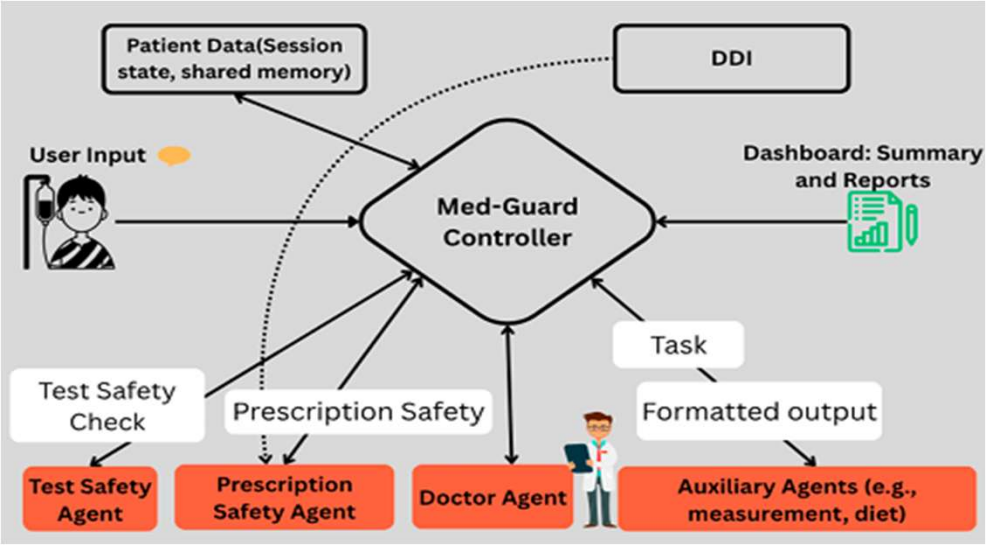
- Diagnostic error contributes to adverse clinical outcomes in healthcare environments.
- Existing LLM based systems face safety risks, hallucinations, and lack of transparency.
- Clinical deployment requires safety verification, auditability, transparency, and structured decision workflows.

Key Idea

- Design a multi-agent LLM framework with explicit safety guardrails.
- Distribute clinical reasoning across specialized agents with defined roles.
- Introduce mandatory safety checks before diagnostic or treatment decisions.
- Introduce differential diagnosis instead of providing single diagnosis.

Contribution

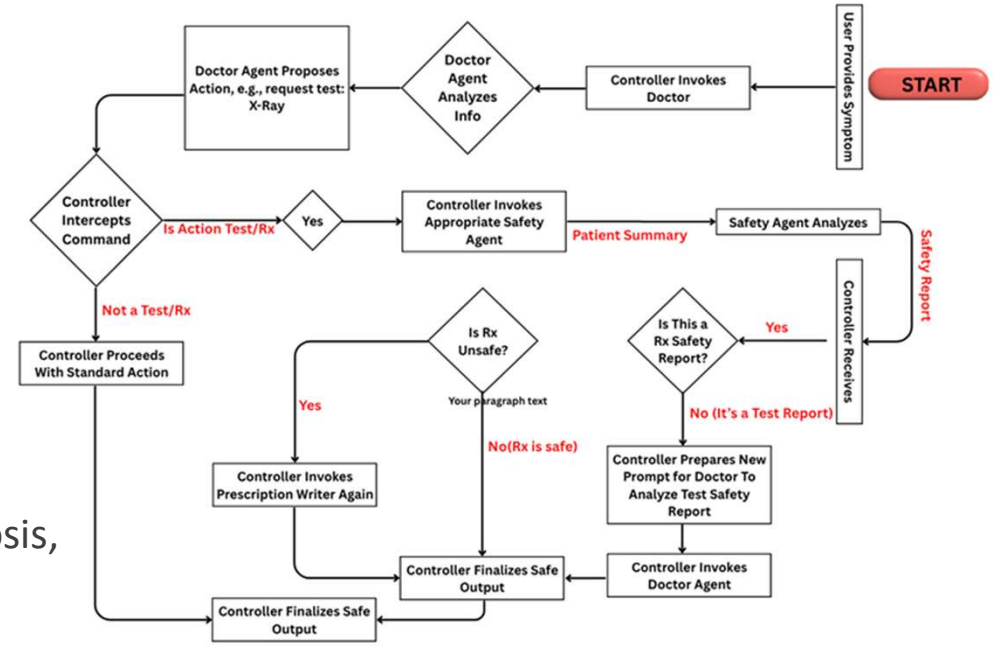
- Safety-first multi-agent architecture for clinical decisions.
- Dedicated safety agents for diagnostic and prescription validation.
- Structured logging and traceability of clinical reasoning steps.



Med-Guard Architecture And Agent Roles

System Design

- Med-Guard uses multiple specialized agents instead of one model.
- A central controller manages workflow and communication.
- Each agent performs a specific clinical task.
- Safety checks are applied before decisions reach the user



Agent Roles

- **Doctor Agent:** collects symptoms, reasons about diagnosis, suggests tests and treatment.
- **Test Safety Agent:** checks risks of diagnostic tests.
- **Prescription Writer Agent:** drafts medication plan.
- **Prescription Safety Agent:** checks drug interactions, allergies, and contraindications.
- **Measurement Agent:** converts lab results into structured formats.
- **Dietary Advisor Agent:** provides lifestyle advice after

Med-Guard Safety Agent And Evaluation Results

Evaluation Setup

- Evaluated on AgentClinic-MedQA diagnostic dataset
- Dataset contains clinical cases with symptoms, history, and lab results.
- System tested on diagnostic accuracy and safety.

Safety Agent Design

- Safety agents verify decisions before they reach the user.
- Independently reviews diagnostic tests and prescriptions.
- Checks for drug-drug interactions.
- Identifies allergy conflicts and contraindications.
- Flags unsafe or high-risk recommendations.
- Prevents unsafe outputs from reaching the patient.

Diagnostic Performance

- **Differential Diagnosis:**
  - Doctor Agent generates a list of candidate diseases.
  - Diagnosis is updated iteratively before final decision.
- Med-Guard **improves diagnostic accuracy** compared to AgentClinic, attributed to differential diagnosis.
- **Example:** Llama-4-Maveric: 66.8% (Med-Guard) vs 42.4% (baseline).

Quantitative Impact

Diagnostic Models	Harmful Prescriptions Detected Successfully
Med-Guard w/o Safety-Agent	44.5
Med-Guard w/ Safety-Agent	71.7

TABLE II

EVALUATION OF THE EFFECTIVENESS OF OUR PRESCRIPTION SAFETY AGENT IN IDENTIFYING AND MITIGATING HARMFUL PRESCRIPTIONS.

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Trust-X: Towards Transparent And Safe Multi-Agent Reasoning

Problem

- Models can be accurate but still reason unreliably.
- Correct answers do not guarantee safe or coherent reasoning.
- Traditional systems lack visibility into intermediate steps.

Contribution

- Introduces Consensus Disagreement Rate (*CDR)* among multiple doctor agents.
- Proposes Epistemic Trust Index (ETI)
- Defines Operational Safety Index (OSI).
- Introduces Final Trust Index (FTI) combining reasoning and safety.

Key Idea

- **Consensus Reasoning:** multiple doctor agents analyze the case independently and vote on the best diagnosis. Their disagreement is recorded as consensus disagreement rate (*CDR*).
- **Reasoning Trace:** each diagnosis step includes a clear trace linking evidence, and reasoning showing how thoughts evolved.
- **End-to-End Logging:** every agent action is recorded with time role and reason, allowing full replay and accountability.
- **Trust Scores:** a trust layer combines interpretability, safety and reasoning quality into measurable trust scores.

Trust-X: Methodology

Multi-Agent Reasoning

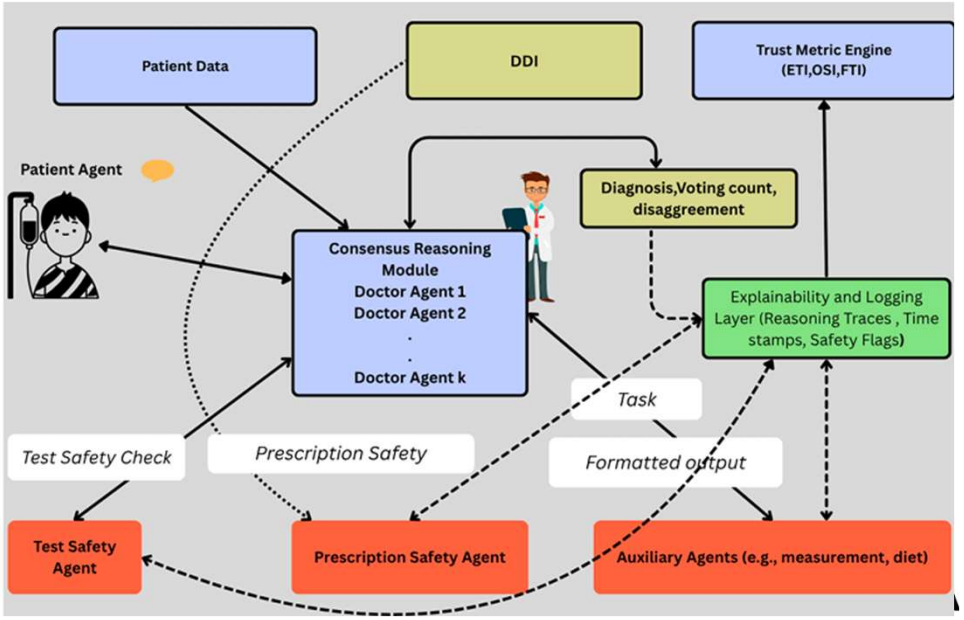
- Three Doctor Agents (K = 3) analyze the same patient case.
- Each produces structured reasoning steps using Differential Diagnosis Lifecycle (DDxL).
- Intermediate hypothesis and discarded diagnoses are logged

Consensus and Uncertainty

- Outputs compared across agents.
- Consensus Disagreement Rate measures variation among diagnoses.
- Higher disagreement indicates higher epistemic uncertainty.
- Alerts and overrides are recorded.

Contribution

- **ETI:** Combines diagnostic accuracy, disagreement, and reasoning coherence.
- **OSI:** penalizes unsafe prescriptions and missed alerts.
- **FTI:** weighted combination of ETI and OSI.



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Trust-X: Trust Metrics And Model Comparison

Evaluation Setup

- 50 MedQA clinical cases.
- **Configurations:**
  - **Baseline:** single doctor agent reasoning without consensus or safety checks.
  - **Consensus:** multiple doctor agents reasons. Final decision based on agreement between agents.
  - **Safety:** safety agents are enabled
  - **Trust:** combines consensus reasoning and safety monitoring.

Accuracy

- Accuracy remains similar (~68-69%) across systems, but trust differs sharply when safety supervision is enabled.

Trust Results

- **Baseline:**
  - OSI = 0 (no safety agents)
  - FTI = 39.2
- **Consensus:**
  - OSI = 0 (no safety agents)
  - FTI = 35
- **Safety:**
  - FTI = 75
- **Trust:**
  - FTI = 75

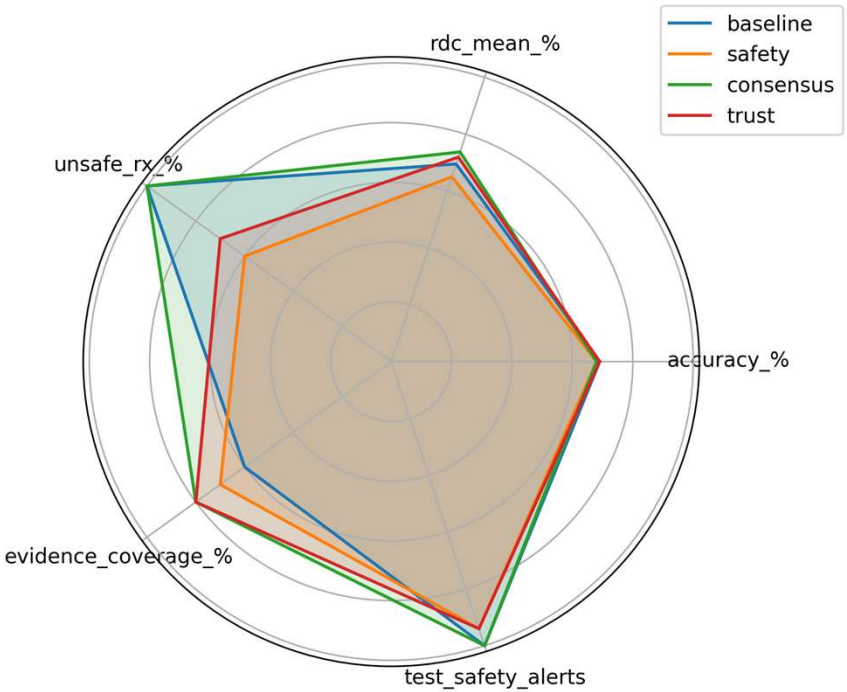
Trust-X: Multidimensional Trust Profile

Multidimensional Evaluation

- Trust evaluated across multiple behavioral dimensions.
- Metrics include diagnostic accuracy, reasoning coherence, evidence coverage, and safety behavior.
- Safety dimension computed using UnsafeRx and TesAlerts.

Observed Patterns

- **Baseline:** fluent but overconfident reasoning with no explicit safety awareness.
- **Consensus:** improved reflection but limited operational safety.
- **Safety:** cautious reasoning and strong risk detection.
- **Trust:** most balanced profile combining reasoning stability and safety oversight.



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